

The BGM Model, Formally

13.1 Setting up the full curve

Fix the tenor structure $0 = T_0 < T_1 < \dots < T_N$ from Chapter 11, with $\tau_i = T_{i+1} - T_i$, and forward rates $L_i(t) = L(t; T_i, T_{i+1})$ for $i = 0, \dots, N-1$. Each L_i is driven by a (possibly multi-dimensional) Brownian motion, with instantaneous correlation between rates i and j written ρ_{ij} — Chapter 14 discusses how this correlation structure is chosen in practice; for this chapter, take it as given. Chapter 11 postulated that, under L_i 's own forward measure $\mathbb{Q}^{i+1}$,

$$dL_i(t) = \sigma_i(t) L_i(t) dW_i^{i+1}(t).$$

The task of this chapter is to find the drift L_i must carry under *any other* measure, so that a whole curve of forward rates can be evolved jointly and consistently.

13.2 The building block: one step of numeraire

Girsanov (Chapter 12) says a drift correction equals the volatility of the numeraire ratio connecting the two measures. The natural building block, therefore, is the numeraire ratio between two *adjacent* bonds, $P(t, T_j)/P(t, T_{j+1})$, since this is precisely $1 + \tau_j L_j(t)$ — already expressed in terms of a forward rate we understand.

Key result: the volatility of one numeraire step

Since $1 + \tau_j L_j(t) = P(t, T_j)/P(t, T_{j+1})$ and $dL_j = \sigma_j L_j dW_j^{j+1}$ under $\mathbb{Q}^{j+1}$, an application of Itô's lemma to $\log(1 + \tau_j L_j(t))$ shows that the volatility of $P(t, T_j)/P(t, T_{j+1})$ — the quantity Girsanov needs — is

$$\gamma_j(t) = \frac{\tau_j \sigma_j(t) L_j(t)}{1 + \tau_j L_j(t)}.$$

This single expression, appropriately summed and signed, is the entire content of the BGM drift correction: moving L_i 's measure past bond T_j costs a drift term of size $\sigma_i \rho_{ij} \gamma_j$.

13.3 The terminal measure

The cleanest common measure to work under is the **terminal measure** $\mathbb{Q}^N$, associated with numeraire $P(t, T_N)$ — the very last bond in the tenor structure. To move L_i from its own measure $\mathbb{Q}^{i+1}$ to $\mathbb{Q}^N$, Girsanov must be applied once for every intermediate bond ratio between T_{i+1} and T_N , and the drift corrections accumulate additively.

Key result: the BGM drift under the terminal measure

Under $\mathbb{Q}^N$, for $i < N-1$:

$$dL_i(t) = -\sigma_i(t) L_i(t) \sum_{j=i+1}^{N-1} \frac{\tau_j \rho_{ij} \sigma_j(t) L_j(t)}{1 + \tau_j L_j(t)} dt + \sigma_i(t) L_i(t) dW_i^N(t)$$

The final rate, $L_{N-1}(t)$, needs no correction at all: T_N is its own natural payment date, so its own measure and the terminal measure coincide, and it remains exactly driftless.

The structure of this formula rewards a close look. Each term in the sum is a product of three things: rate i 's own volatility σ_i (shared by every term, factored out), rate j 's volatility σ_j , and the correlation ρ_{ij} between them, all scaled by the $\tau_j L_j / (1 + \tau_j L_j)$ factor from the building block above. This factor is always less than 1 and grows with the rate level L_j — so the drift correction is larger when rates are higher, a direct, mechanical consequence of exactly how much a bond-price ratio moves for a given move in the underlying forward rate.

PITFALL

The drift sum runs from $j = i + 1$ to $N - 1$ — *forward* in tenor from rate i to the end of the curve, never backward. It is tempting, when implementing this for the first time, to sum over all $j \neq i$ symmetrically, as if every rate's drift depended on every other rate's volatility equally in both directions. It does not: under the *terminal* measure specifically, only rates with a later reset date than i contribute to i 's drift. Get the summation range wrong and the model still runs, still produces plausible-looking numbers, and is silently arbitrageable — a serious defect that will not show up as a crash and may not show up as an obviously wrong price on any single instrument, only as a subtle failure of internal consistency across instruments.

Worked example 13.1 — computing one drift term by hand.

Consider a three-rate BGM system with $\tau = 0.5$ throughout, terminal measure $\mathbb{Q}^3$ (so $N = 3$, and L_0, L_1 need drift corrections while L_2 does not). At some time t : $L_1(t) = 4.20\%$, $L_2(t) = 4.35\%$, $\sigma_1 = 22\%$, $\sigma_2 = 24\%$ (lognormal vols), and $\rho_{12} = 0.85$. Compute the drift correction on L_1 coming from the $j = 2$ term only.

Step 1 — the numeraire-step volatility γ_2 .

$$\gamma_2 = \frac{\tau \sigma_2 L_2}{1 + \tau L_2} = \frac{0.5 \times 0.24 \times 0.0435}{1 + 0.5 \times 0.0435} = \frac{0.005220}{1.02175} = 0.005110$$

Step 2 — the drift contribution.

$$\text{drift}_{1 \leftarrow 2} = -\sigma_1 L_1 \times \rho_{12} \times \gamma_2 = -0.22 \times 0.0420 \times 0.85 \times 0.005110$$

$$-0.22 \times 0.0420 = -0.009240$$

$$-0.009240 \times 0.85 \times 0.005110 = -4.014 \times 10^{-5}$$

Sanity check. The result is a small negative number, of order 10^{-5} per unit time — and it should be small: it represents a correction to an annual percentage drift, and $\gamma_2 \approx 0.5\%$ being multiplied through two more sub-1 factors necessarily produces something an order of magnitude smaller again. A drift correction of order 1 (i.e. 100% per year) would indicate a units error somewhere in the pipeline, not a genuine result.

Interpretation. Over a short time step Δt , this term contributes roughly $-4.0 \times 10^{-5} \times \Delta t$ to dL_1/L_1 — a genuinely small effect for a single term, but a full drift sums many such terms across every later rate in the curve, and the cumulative effect across a 30-year, quarterly-tenor curve (roughly 120 terms in the longest sum) is far from negligible. This is exactly why the drift cannot be approximated away in a serious implementation, and why Chapter 19's numerical schemes are built specifically around computing it efficiently.

13.4 The spot measure: the practitioner's alternative

The terminal measure is elegant to state but has a practical drawback: every drift, for every rate, depends on rates *later* in the curve, so pricing anything that involves early cash flows still requires the entire curve to be simulated out to T_N . The **spot LIBOR measure**, associated with a discretely-rolled money-market-account numeraire, fixes this at the cost of a slightly less elegant-looking formula.

Key result: the BGM drift under the spot measure

Let $\eta(t)$ denote the index of the next tenor date after t (so $\eta(t)$ increases as t advances). Under the spot measure:

$$dL_i(t) = \sigma_i(t) L_i(t) \sum_{j=\eta(t)}^i \frac{\tau_j \rho_{ij} \sigma_j(t) L_j(t)}{1 + \tau_j L_j(t)} dt + \sigma_i(t) L_i(t) dW_i^{\text{spot}}(t)$$

Here the sum runs *backward* from i to the nearest already-passed reset date $\eta(t)$, rather than forward to the end of the curve, and the sign is positive rather than negative. Crucially, this means L_i 's drift, under the spot measure, depends only on rates *earlier* in the curve than i itself — which makes a step-by-step Monte Carlo simulation, evolving the whole curve together one time step at a time, considerably more natural to implement than under the terminal measure.

ON THE DESK

In practice, most production BGM Monte Carlo engines use the spot measure specifically because of this forward-in-time-only dependency structure — it maps naturally onto a standard “evolve the whole curve one time step, then the next, then the next” simulation loop, whereas the terminal measure's backward-looking drift sum sits awkwardly against that loop structure (though it is occasionally preferred for exotics whose payoff is naturally expressed at the terminal date, where its driftlessness for the last rate becomes a genuine convenience rather than a liability). Neither measure is “more correct” — they price identically, being related by exactly the change-of-numeraire machinery of Chapter 12 — the choice is purely one of numerical and implementation convenience.

13.5 The general drift, and where displacement fits

Both formulas above are special cases of a single general pattern: the drift of L_i under any measure equals $\sigma_i L_i$ times a signed sum of $\rho_{ij} \gamma_j$ terms, over whichever range of j connects i 's own measure to the working measure. Chapter 15 will modify the volatility term itself — replacing $\sigma_i(t) L_i(t)$ with $\sigma_i(t)(L_i(t) + c_i)$ for a displacement constant c_i , to handle low or negative rates — but the drift-derivation machinery of this chapter carries over essentially unchanged, with L_j replaced by $L_j + c_j$ throughout. It is worth internalising this chapter's derivation precisely because so much of what follows is a direct reuse of it, not a fresh derivation.

Chapter benchmark

Before moving on, you should be able to, without notes:

1. Derive the volatility of a single bond-price-ratio numeraire step, $\gamma_j(t) = \tau_j \sigma_j L_j / (1 + \tau_j L_j)$, from L_j 's own driftless dynamics.

2. State the BGM drift formula under the terminal measure, and explain why the summation range runs forward from $i + 1$ to $N - 1$.
3. State the BGM drift formula under the spot measure, explain why its summation range runs backward instead, and explain why this makes it more convenient for a step-by-step Monte Carlo implementation.